

CD44 promoter-reporter architectures

5' -> 3' display, genomic 3513721 -> 35233402 | strand + | assembly GRCh38 | annotation Ensembl live GRCh38; selected reporter transcripts | upstream 5000 bp | downstream 1000 bp

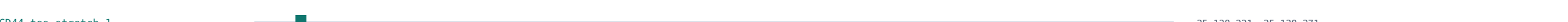
Canonical reporter comparison: 2 TSS class(es), 3 transcript(s), 9 explicit architecture row(s); normalized evidence remains observational/predictive context.



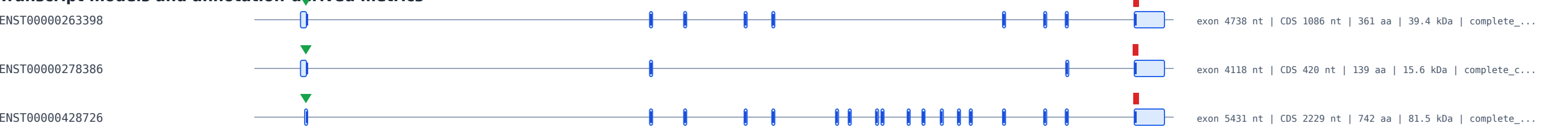
Legend (annotation-derived transcript features)



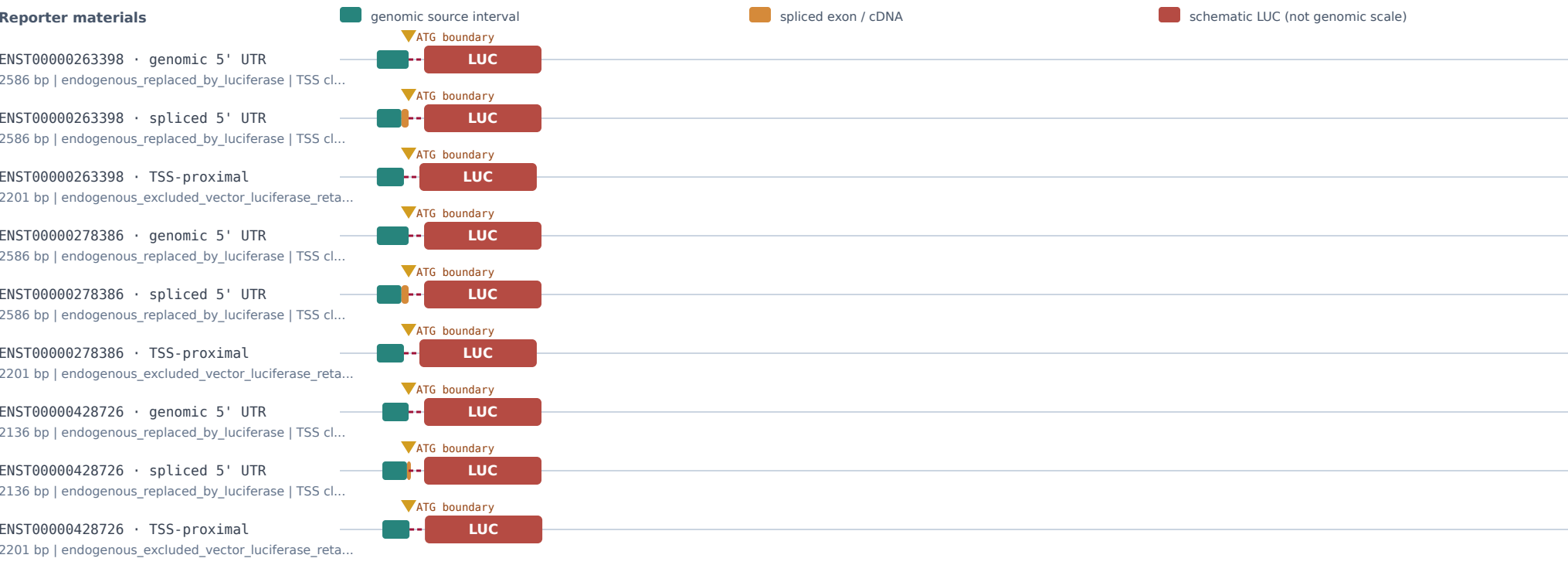
TSS stretches: reference for similarity below



Transcript models and annotation-derived metrics

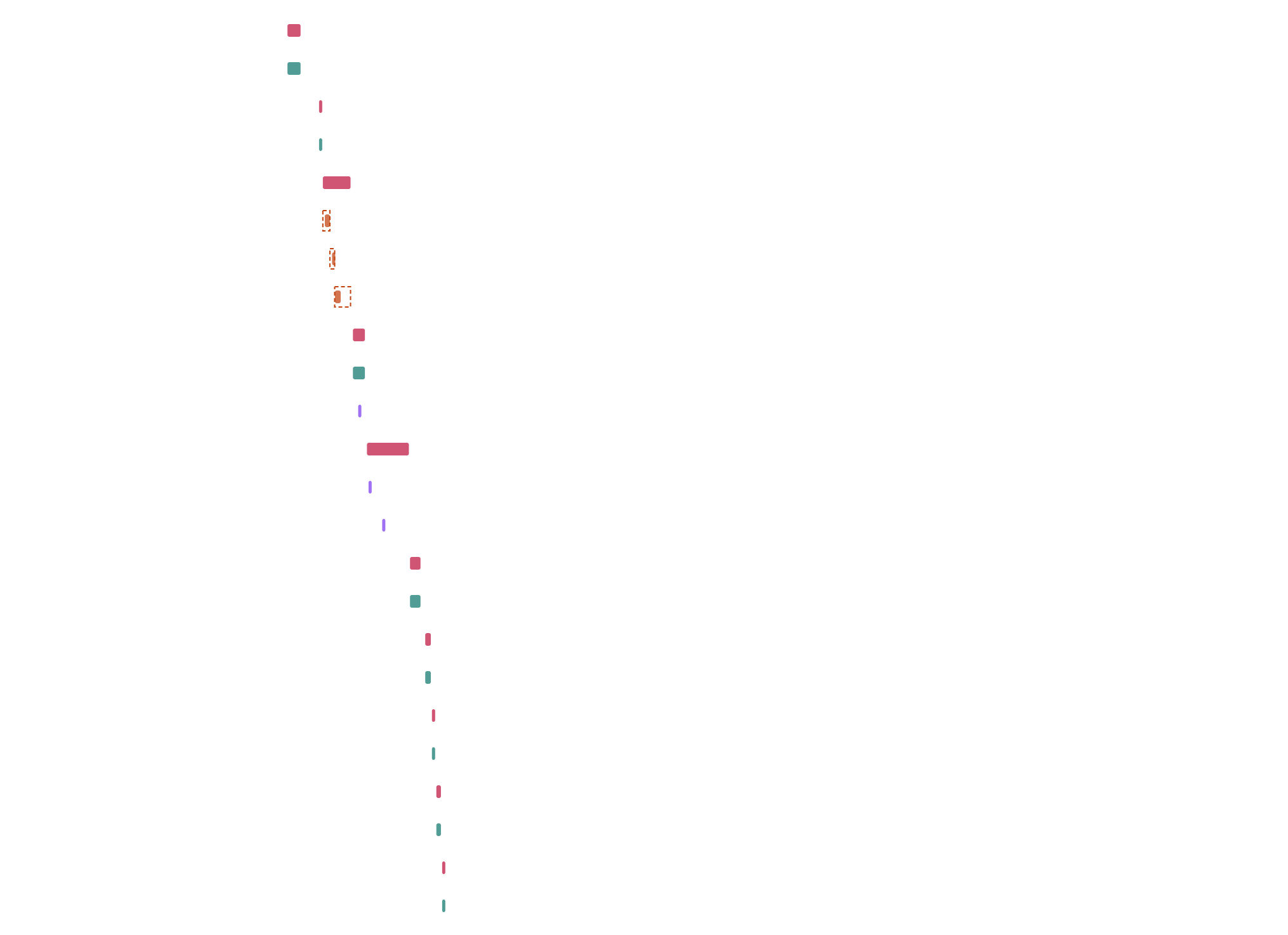


Proposed reporter architectures



Ensembl-annotated regulatory regions

Ensembl Regulation | release 2026-08 | pipeline 2.1 | GRCh38



64 additional source-bound row(s) omitted from SVG; complete rows remain in JSON.

Projected occupancy grouped by declared biological context

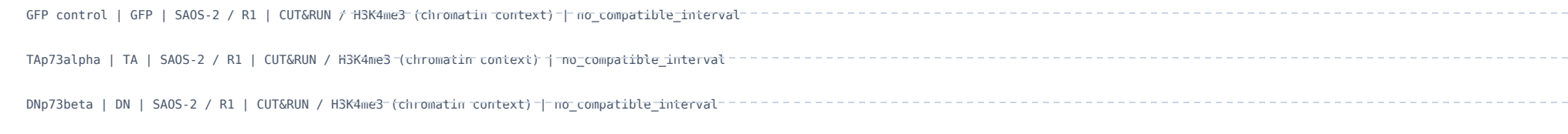
TP73 occupancy — SA05-2 | shared within group | observed group max 1.000



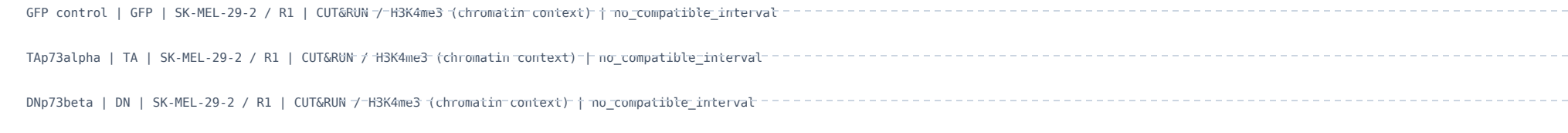
TP73 occupancy — SK-MEL-29-2 | shared within group | observed group max 1.000



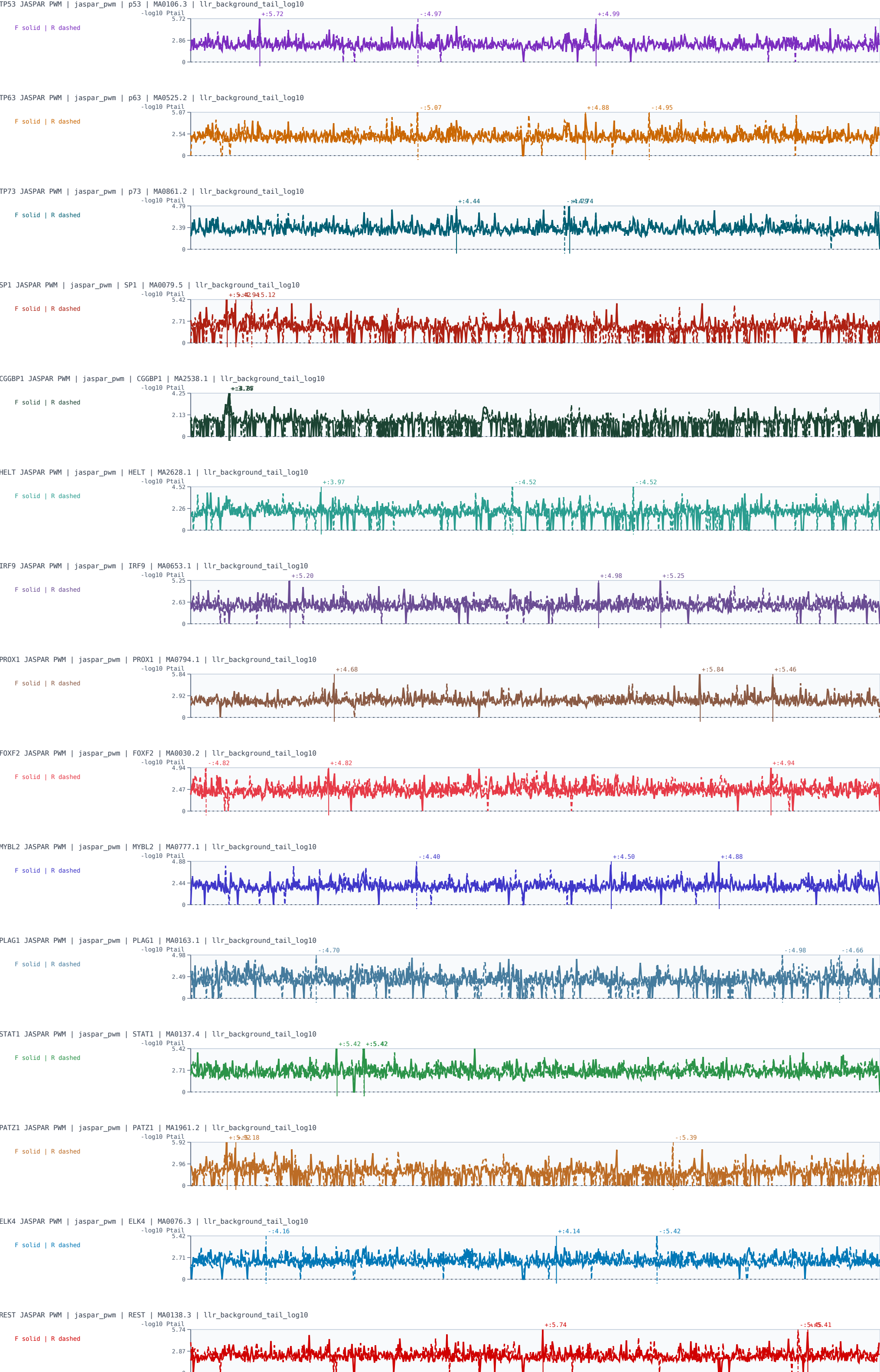
H3K4me3 chromatin context — SA05-2 | shared within group | observed group max 1.000



H3K4me3 chromatin context — SK-MEL-29-2 | shared within group | observed group max 1.000



Predicted TF binding scores



TSS-local Ensembl-feature promoterome similarity

Transcript overlap ~500/+200 bp; features clipped to the displayed stretch; BLASTN ≥40 bp, ≥80% identity, E=1e-05; self-locus and same-gene targets excluded.

Blue frequency strips count distinct other genes by query position. Numbered blocks give target-promoter order; red outlines mark order/orientation breaks.



Similarity is structural evidence; Ensembl classes, predicted TFBS and CUT&RUN enrichment do not prove reporter activity or sufficiency.

Reporter interpretation boundaries

Transcript annotation alone does not establish which TSS is used in the assayed cells.

CUT&RUN overlap is occupancy/enrichment evidence, not proof of TSS usage, promoter activity, direct regulation, biochemical affinity, or luciferase output.

Reporter geometry does not verify the identity of a physical laboratory construct.

Warnings / interpretation checks

warning: Occupancy lane 'SA05-2 H3K4me3 Dnp73beta' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SA05-2 H3K4me3 GFP' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SA05-2 H3K4me3 Tap73alpha' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SA05-2 TP73 Dnp73beta' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SA05-2 TP73 GFP' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SA05-2 TP73 Tap73alpha' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SK-MEL-29-2 H3K4me3 Dnp73beta' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SK-MEL-29-2 H3K4me3 GFP' is retained as no_compatible_interval; no compatible interval is rendered.

Evidence provenance

ensembl_regulation_annotation: ensembl_regulation_2026_08_grch38

ensembl_regulation_annotation_intervals: ensembl_regulation_2026_08_grch38

isoform_panel: CD44_panel

occupancy_layout: tp73_sa052,tp73_skme129_2,h3k4me3_sa052,h3k4me3_skme129_2

predicted_tf_score_jaspar_pwm: local_jaspar_registry

requested_occupancy_track: E-MTAB-15709:h3k4me3:sa052:DN:R1

requested_occupancy_track: E-MTAB-15709:h3k4me3:sa052:GFP:R1

requested_occupancy_track: E-MTAB-15709:h3k4me3:sa052:TA:R1

requested_occupancy_track: E-MTAB-15709:h3k4me3:skme129-2:DN:R1

requested_occupancy_track: E-MTAB-15709:h3k4me3:skme129-2:GFP:R1